Rethinking the Arithmetic of Multi-Agent Medical Consultation

Anonymous ACL submission

Abstract

Medical language-model systems use multi-agent consultation, prompting agents into clinical specialties to discuss cases and reach a shared answer. This design relies on specialists making different mistakes. In this work, we measure their error correlation across 480 multi-agent configurations and 161,999 clinical episodes. At nine members, an Independent panel is unanimous on 67.5% of items and wrong on 38.1% of what it agrees on. Errors correlate at $\varphi = 0.76$, giving nine agents 1.27 effective independent opinions, falling to 1.01 after discussion. Mixing models from six vendors yields 2.29 effective opinions. Two published medical agent frameworks, run from their own code, have error correlations of 0.860 and 0.863. In simpler terms: a panel of nine is worth one, and it agrees anyway; its agreement must not reach a clinician as confidence.

1 Introduction

Medicine has a procedure for a hard case: convene several specialists, let each form a view, and see whether the views agree. Clinicians trained differently make different mistakes, so agreement on a diagnosis provides corroborating evidence. That is why a tumour board's agreement informs a treatment decision and why a matching second opinion supports the initial diagnosis.

Multi-agent consultation transplants the procedure into language-model systems. MedAgents (Tang et al., 2024), MDAgents (Kim et al., 2024), MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) build the same object: several agents prompted into clinical specialties, discussing cases until they agree, on the reasoning that a panel of specialists diagnoses better than the single doctor. This reasoning requires the specialists to make different mistakes: pooling correlated judgements repeats them rather than combining them (Kurvers et al., 2016; Kämmer et al., 2017). Figure 1 contrasts this independence premise with the behaviour of role-prompted panels.

In this work we measure it. A consultation is worth assembling only if somebody in the room is right when the others are wrong, and the room can tell who. We measure headroom and capture rate κ; collaboration gain is exactly their product. Our evaluation covers 480 multi-agent configurations and 161,999 episodes. It uses eight models from four vendors, three medical benchmarks, four multi-agent architectures and panel sizes $N \in \{1, 3, 5, 7, 9\}$. A budget-matched single-model control accompanies every cell.

Agents prompted into different medical specialties fail on the same items. Their pairwise error correlation is $\varphi = 0.76$, leaving a nine-member panel with 1.27 effective independent opinions. Peer discussion raises the correlation to 0.990, leaving 1.01 effective opinions. Computed from raw φ, mixing model generations within one vendor yields 2.60 effective opinions in a nine-member panel; mixing models from six vendors yields 2.29.

Run from their own repositories on the same items with `gpt-4.1-nano`, MedAgents (Tang et al., 2024) and MDAgents (Kim et al., 2024) gain $+5.7$ and $+1.2$ percentage points over the single doctor, a factor of five apart. Their agents' errors correlate at 0.860 and 0.863, above 0.792 for our panels on the same model.

Selection limits the gain from this headroom. At nine agents the oracle ceiling is 63.9% against 53.1% for the single doctor, giving 10.7 pp of headroom. Across the four architectures and self-consistency, capture rate κ ranges from -7.4% for Hybrid to 17.7% for Decentralized. Two classifiers trained on fifteen first-round features against ground truth do not outperform plurality voting.

In medicine, agreement affects safety. At $N{=}9$, discussion raises unanimity from 67.5% (Indepen-

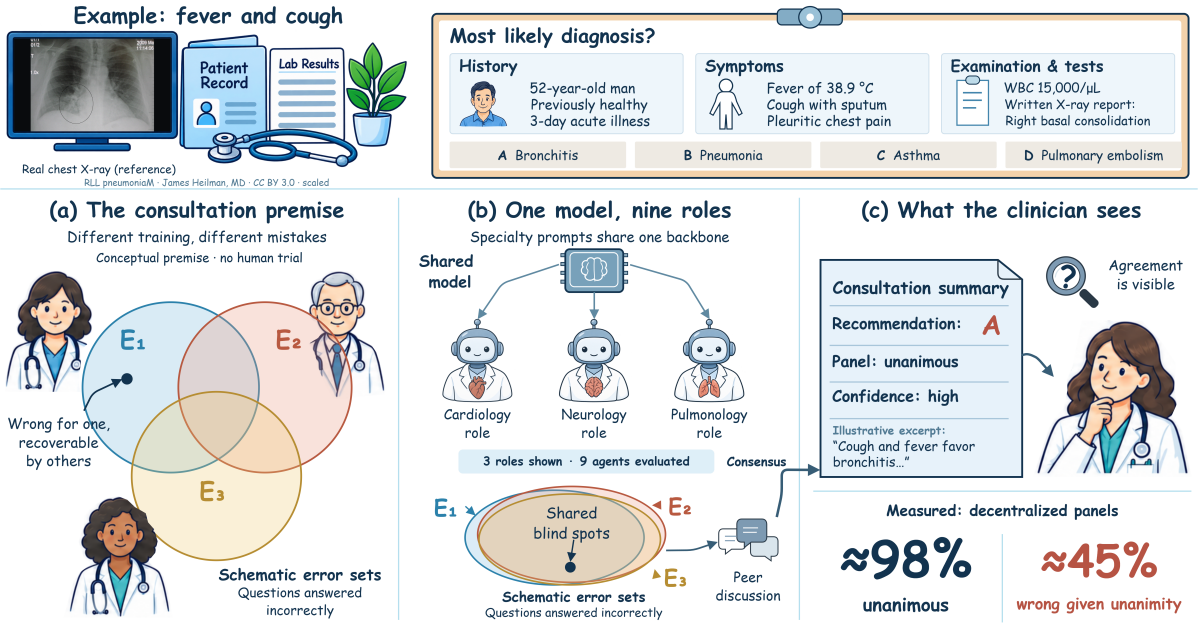


Figure 1: **A panel that agrees is not a panel that knows.** The top row illustrates a case; B is correct. (a) Differently trained clinicians correct each other's errors. (b) Role-prompted agents' shared blind spots become consensus through discussion. (c) Clinicians receive a unanimous, high-confidence wrong recommendation. Decentralized panels (N=9) are 98.0% unanimous; 44.9% of unanimous answers are wrong (Table 6).

dent) to 98.0% (Decentralized), while the error rate among unanimous answers rises from 38.1% to 44.9%. Unanimity separates the panel's right answers from its wrong ones at AUC = 0.59.

The rest of the paper is organised as follows. Section 2 places the work against the collective-intelligence results the design pattern borrows from and the medical systems built on them. Section 3.1 defines the two measurements and the seven conditions. Section 4.3 examines error correlation and the four sources of panel diversity; Section 4.4 measures how much headroom aggregation recovers; and Section 4.6 reports the effects of consultation on agreement.

2 Related work

Why pooled judgements help, and what they require. Collective-intelligence results in medicine rest on error independence: pooling independent diagnostic judgements improves accuracy (Kurvers et al., 2016; Wolf et al., 2015), plateauing at roughly five raters (Kämmer et al., 2017), and groups of nine outperform individuals on the Human Dx platform (Barnett et al., 2019), the clinical instance of a general principle (Surowiecki, 2004) that multidisciplinary tumour boards institutionalise (Radcliffe et al., 2019). Hautz and Kämmer (2024) ask when two heads beat one. The same literature documents what happens when independence fails: conformity to a unanimous majority (Asch, 1956) and groupthink in deliberating groups (Janis, 1972). Correlated errors across models bound ensemble benefit in general (Kim et al., 2025), and multi-agent debate can converge on confident-but-wrong answers (Zhu et al., 2026). Diagnostic error is common in ordinary practice (Graber, 2013; Singh et al., 2014), making agreement on an incorrect answer a safety concern. Pooling 40,762 physician differentials with five frontier models over 2,133 open-ended vignettes, Zöller et al. (2025) find that human–model collectives beat physician collectives and model ensembles. They attribute the margin to humans and models making different kinds of errors.

Medical language models and multi-agent systems. Medical question answering became a capability benchmark once models matched licensing-exam thresholds (Kung et al., 2023; Nori et al., 2023; Singhal et al., 2023, 2025), and later work moved to conducting encounters (Tu et al., 2025; McDuff et al., 2025). Prompting strategy carries much of that progress (Nori et al., 2024), which is why architectural claims need budget-matched controls. On the multi-agent side, MedAgents (Tang et al., 2024) introduced role-played specialist collaboration, MDAgents (Kim et al.,

2024) added adaptive routing, and MAC (Chen et al., 2025) and TeamMedAgents (Mishra et al., 2026) formalised teamwork components, the latter sweeping team sizes 2–5. The medical ablations cited here stop at $N \leq 5$ and omit panel-size curves and budget-matched single-model controls. Sceptical evaluations find that orchestration often fails to beat well-prompted single models at substantially higher cost (Zhu et al., 2025; Liu et al., 2026; Zhao et al., 2025). Chen et al. (2024b) and Zhou et al. (2025) specialise the role prompts, using an MDT for rare disease and role-assigned radiologist and generalist agents for multimodal input. Liu et al. (2025) adds a director to coordinate them. Ma et al. (2025) replaces majority voting with logical contradiction resolution and Li et al. (2025) lets agents accumulate diagnostic knowledge across a cohort, both of which are attempts to escape a voting rule rather than to escape the correlation that makes voting weak. Fan et al. (2024), Sanghvi et al. (2026) and Sviridov et al. (2025) move the setting itself, from single-shot selection to simulated, interactive and multimodal encounters where agents must ask before they answer, and Chen et al. (2026) and Shi et al. (2026) carry the pattern into inpatient pathways and outcome prediction, where triage, diagnosis and treatment are separate coordinated stages. These systems vary the aggregation rule, specialty roles or interaction protocol. Capture rate κ, measured in §4.4, evaluates how much headroom an aggregation rule recovers.

Closest to our central question, Yuan et al. (2026) compare single-vendor and mixed-vendor panels of three frontier models on rare-disease diagnosis and report that mixed-vendor teams win, attributing the gain to complementary inductive biases surfaced by an overlap analysis. Table 4 examines the same source of diversity using error correlation.

Scaling agents, and scaling inference. Debate protocols improve factuality (Du et al., 2024; Liang et al., 2024) and frameworks such as MetaGPT (Hong et al., 2024) and AutoGen (Wu et al., 2024) made the pattern easy to deploy (Zhang et al., 2024). On how far it scales, "more agents is all you need" (Li et al., 2024) reported monotone gains, while Chen et al. (2024a) showed the relationship is non-monotone and Li et al. (2026) found an inverted-U in agent count; Yang et al. (2026) attributed the shape to opinion diversity, showing two diverse agents can match sixteen homogeneous ones. Kim et al. (2026) fit a predictive scaling principle across architectures, model families and agentic benchmarks. Scaling laws describe training compute (Kaplan et al., 2020; Hoffmann et al., 2022). At test time, repeated sampling (Brown et al., 2024) and compute-optimal inference (Snell et al., 2024) offer alternatives to adding agents, and self-consistency (Wang et al., 2023) over chain-of-thought (Wei et al., 2022) is the budget-matched control we run at every cell.

Clinical reliability. Work on model self-knowledge (Kadavath et al., 2022) and confidence elicitation (Xiong et al., 2024; Guo et al., 2017) examines how models report uncertainty. Clinical evaluations report fluent but unreliable responses (Goodman et al., 2023) with demographic biases that propagate (Omiye et al., 2023; Zack et al., 2024).

3 Measuring what a consultation adds

3.1 The measurement problem

Complementary and redundant panels produce the same observable: several agents, an aggregation step, one answer. Accuracy cannot separate them. A panel can gain a point because its members caught each other's errors, or because nine correlated draws from one distribution landed on the mode more often than one draw did. This is why the multi-agent medical literature can report gains and the sceptical literature can report that orchestration fails to beat a well-prompted single model (Zhu et al., 2025; Liu et al., 2026; Zhao et al., 2025) without either side being wrong.

The premise the design pattern inherits is specific enough to test. Pooling judgements improves accuracy when the judgements are independent; correlated judgements are merely repeated. So we instrument each episode with the two quantities that premise depends on, and report them alongside accuracy throughout.

How many opinions a panel contains. Writing φ for the mean pairwise error correlation between panel positions (the correlation of their per-item error indicators), the effective number of independent opinions in a panel of N agents, with $n_a \equiv N$, is

$$N_{\text{eff}} = \frac{N}{1 + (N-1)\varphi}. \quad (1)$$

At $\varphi = 0$ the N opinions are independent and $N_{\text{eff}} = N$; at $\varphi = 1$ they are one opinion copied N times and $N_{\text{eff}} = 1$. This is the design-effect correction for clustered observations. Applied to a panel of agents, it converts nominal panel size into the number of effective independent opinions.

Raw φ depends on the members' error rates. Two binary error indicators with unequal marginals cannot reach $\varphi = 1$: with error rates $p \leq q$ the attainable maximum is

$$\varphi_{\max} = \sqrt{\frac{p\,(1-q)}{q\,(1-p)}}, \tag{2}$$

which falls as the members' accuracies diverge. A panel assembled from models of different strength therefore posts a lower raw φ whether or not its members reason differently. We report $\varphi/\varphi_{\max}$ wherever members differ in accuracy.

What a panel holds, and what its aggregation keeps. Independence sets headroom; an architecture determines panel accuracy. Let the *oracle ceiling* P_{oracle} be the accuracy of a rule that selects a correct first-round opinion whenever any member holds one. It bounds aggregation rules that select from those opinions without seeing the label. Collaboration gain then factors exactly:

$$\underbrace{P - P_{\text{SA}}}_{\text{gain}} = \underbrace{(P_{\text{oracle}} - P_{\text{SA}})}_{\text{headroom}} \times \underbrace{\kappa}_{\text{capture}}, \tag{3}$$

where P_{SA} is the single-doctor baseline on the same items and the capture rate κ is the fraction of headroom an architecture recovers. Here $\kappa = 0$ is the single-doctor baseline and $\kappa = 100\%$ is the oracle. A negative κ means panel accuracy is below the single-doctor baseline. The factorisation is definitional. The two factors have different causes and remedies. Headroom depends on independent opinions; κ depends on selecting a correct opinion among members who disagree.

We also record *error amplification* A_e: how often the panel held the correct option among its first opinions and output a wrong one, normalised by the marginal probability that a single agent is wrong. This differs from Kim et al. (2026)'s $A_e = E_{\text{MAS}}/E_{\text{SAS}}$.

3.2 Conditions

An agent system is a triple (A, C, Ω): agents, a communication graph, and an aggregation rule (Fig. 2). The taxonomy is the one used for general-domain agent systems (Kim et al., 2026); the clinical instantiations are ours. All conditions share the item rendering and answer schema. Specialist panels share specialty role prompts; SELF-CONSISTENCY and GENERIC PANEL use the generic prompt. Panel members and self-consistency samples run at temperature 0.7, while single-doctor baselines and orchestrator/triage turns run at 0.3. Reasoning models ignore temperature.

SINGLE. One agent answers, zero-shot or with chain-of-thought. It is the baseline P_{SA} against which every gain in this paper is measured, run on the same items in the same cell as the panels it is compared to.

SELF-CONSISTENCY. The same model, the same *generic* prompt, k samples at the panel-member temperature, aggregated by majority vote at a matched dollar budget. Comparing it to a specialist panel measures what specialty role prompts contribute over sampling the same model repeatedly.

INDEPENDENT. N specialists receive the item and answer without seeing one another; aggregation is majority vote, ties broken by mean stated confidence and then option order. With $C = \emptyset$, specialty role prompts and sampling produce disagreement. This condition measures role prompting without communication.

CENTRALIZED. An orchestrator receives the N opinions on a star topology and must *audit before aggregating*: name the single opinion most likely to be wrong and justify that, then either commit or re-task one named specialty, which re-answers with the critique in hand. Specialists never see each other, so the star is preserved. The orchestrator plays the attending physician, a designated point at which an error can be caught before it reaches the output. This condition tests whether an orchestrator improves κ.

DECENTRALIZED. N specialists give first opinions, then exchange them on a complete graph for up to three rounds, each free to revise. Rounds stop early only on true unanimity, defined as every agent producing the same *valid* option; a panel containing an unparsed answer is not unanimous. This condition models a multidisciplinary team meeting and measures independence and agreement before and after discussion.

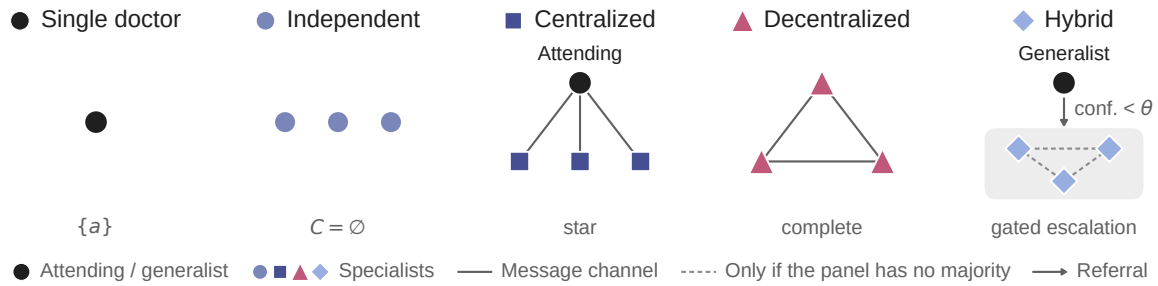


Figure 2: **The coordination structures we evaluate.** Architecture colours and shapes recur throughout. Dark circles mark the single doctor, attending and generalist; coloured markers denote specialists. Solid lines carry messages. Hybrid's generalist refers cases to specialist panels only when confidence is below θ; dashed peer channels open only without a panel majority. SELF-CONSISTENCY and GENERIC PANEL vary no topology and are omitted.

HYBRID. A generalist answers first with a stated confidence; the triage gate escalates the case to the N-specialist panel when that confidence falls below a frozen threshold θ. A panel with no majority escalates to full discussion. θ is calibrated once on a 100-item development slice disjoint from every evaluation set. This is confidence-gated escalation from one generalist to specialists, distinct from the Hybrid topology of Kim et al. (2026) (an orchestrator plus limited peer edges).

GENERIC PANEL. INDEPENDENT, CENTRALIZED and DECENTRALIZED rebuilt out of undifferentiated internists: every member receives the same generic system prompt. The model, temperature, architecture and items match the specialist conditions, so the comparison isolates the contribution of specialty role prompts.

Panel composition and nesting. A cheap router (`gpt-4.1-nano`, temperature 0) ranks the nine most relevant specialties for each item from a fixed 17-specialty list; a panel of size N is the first N of that ranking. Every agent's first opinion is keyed on $(\text{item}, \text{specialty}_i, \text{seed}, i)$ and never on N or on the architecture, so panels are *nested* (the $N{=}9$ panel contains the $N{=}5$ panel), and all four multi-agent conditions reuse the same first opinions. The nesting is a common-random-numbers coupling that fixes the estimand as *accuracy as one routed panel is extended* rather than the expected accuracy of independently redrawn panels.

3.3 Clinical tasks

Three benchmarks, 250 items each, sampled with a fixed seed: MedXpertQA-Text (Zuo et al., 2025) (10 options; stratified by the 12 body systems × 3 task types), MedQA-USMLE (Jin et al., 2021) (4 options), and the contamination-screened hard split of MedAgentsBench (Tang et al., 2025) (894 items excluding the `medqa_5options` subset; the first 250 after stratified shuffling). The endpoint is exact match against the dataset's own gold key. No LLM judge appears anywhere in the evaluation path.

Agent systems on general-domain benchmarks call tools, and tool-call overhead dominates the token accounting and confounds any comparison of coordination structures; a panel that coordinates more also retrieves more, and the two effects are not separable from the output. Here no condition calls a tool, so no tool-call overhead enters the difference between an architecture and the single-doctor baseline on the same item.

4 Experiments and results

4.1 Setup

LLMs and capability scaling. Three OpenAI models form capability tiers T1–T3 for the architecture grid (Table 8; the screening probe over eight candidates is Table 10). The tiers are monotone on all three benchmarks and span single-doctor accuracy from 15.6% to 91.6% (15.6% to 41.6% on MedXpertQA), straddling the 45% threshold reported by Kim et al. (2026). Five further models (Gemini 3.5 Flash Lite and Gemini 3.7 Flash from Google, Claude Haiku 4.5 and Claude Sonnet 5 from Anthropic, and `deepseek-v4-flash` from DeepSeek) run on the same items and architecture grid. Five more (`deepseek-v4-pro`, `qwen3.8-flash` and `qwen3.8-max`, `glm-5.3-flash` and `glm-5.3`) contribute single-doctor baselines on the same items for the diversity analysis of §4.3. Thirteen models across six vendors span $I = 34.0$ to

71.1. Capability-matched cross-vendor pairs include `deepseek-v4-pro` at 59.7 with `gpt-5-mini` at 59.2, and Gemini 3.5 Flash Lite at 50.3 with `gpt-5-nano` at 50.8. A task-grounded capability metric explains more variance than a generic intelligence index (Kim et al., 2026), so we define the capability index I as each model's mean single-doctor chain-of-thought accuracy across the three benchmarks. A cell denotes one model–benchmark pair.

Budget-matched controls. Every configuration is accompanied by three single-model controls: zero-shot, chain-of-thought, and self-consistency at a matched budget. The self-consistency pool is built from cached $n{=}8$ sampling calls (the API maximum), so any k is a prefix of the same samples. Nominal cost is charged as $\lceil k/8 \rceil$ prompts plus k completions, which is what an efficient implementation would pay. Panels are compared against the self-consistency configuration whose mean per-item *dollar* cost is closest, not the one with a matching sample count. Matching on samples silently favours the panel, because a panel pays the prompt once per agent while self-consistency amortises it across a batch.

Cost accounting is *nominal* throughout. Cached calls are charged at list price, so the economics figures report what a configuration would cost to run, not what our cache made it cost.

Statistics. Accuracies are means over a benchmark's 250 items, and summaries across cells or configurations report the mean and standard deviation.

Episodes that fail for infrastructure reasons (rate limits, timeouts) are recorded with an error status and excluded from accuracy. Scoring them as wrong answers would bias call-heavy architectures downward.

4.2 Main results: what a panel adds

Across 480 multi-agent configurations and 161,999 episodes, the best configuration in each cell gains between -2.0 and $+7.6$ pp over the single doctor. Table 1 gives each architecture's gain at its best panel size; Figure 6 in Appendix D plots these gains.

Architectures differ little within most cells. At their best panel sizes, the four architectures differ little within a cell, except on `deepseek-v4-flash`, where Hybrid trails the other three architectures by 4.4–8.8 pp on all three benchmarks. The ordering of topologies (star, complete, tiered, or no edges) carries only weakly across benchmarks. Kendall's τ between the architecture rankings of any two benchmarks averages $+0.333$, and leave-one-benchmark-out cross-validation gives $R^2 = -0.047$. On `gpt-5-nano` / MedXpertQA, Independent peaks at 35.6% at $N{=}5$, while Decentralized reaches 38.0% at $N{=}3$. Gains mostly arrive by $N{=}3$. A panel larger than $N{=}5$ helps in 29 of 96 (cell, architecture) pairs, by at most 2.8 pp (Fig. 8, Fig. 10).

Both factors are small, and they move apart. Eq. 3 separates headroom from the capture rate κ (Fig. 3). Above $P_{\mathrm{SA}} = 25\%$, the two factors move in opposite directions, an empirical result that the factorisation does not require (Table 2). Headroom falls from $+14.3$ pp at $P_{\mathrm{SA}} \in [25, 50)$ to $+4.1$ above 70%, while κ rises from $+6.6\%$ to $+29.5\%$. Gain stays between -0.3 and $+1.2$ pp across the four bands.

The rest of this section measures the two factors in turn. §4.3 measures headroom and evaluates the sources of panel diversity. §4.4 measures the capture rate κ. §4.5 applies the same measurements to two published frameworks. §4.6 reports agreement and its relationship to correctness. The last subsection reports panel costs.

4.3 Is anyone in the room right? The arithmetic premise

The headroom in Eq. 3 measures how much independent judgement a panel contains.

Sampling supplies most of the headroom. At $n_a{=}9$ the single doctor scores 53.1% and the oracle ceiling 63.9%: the panel has 10.7 pp of headroom.

Self-consistency gives one model the same generic prompt nine times at one temperature; a panel gives it nine different specialty prompts. Both produce nine answers, so both admit the same oracle measurement. Nine self-consistency samples put a correct answer on the table 63.5% of the time; nine distinct specialists put one there 63.9% (Fig. 4a). Repeated sampling supplies 10.4 pp (97%) of the 10.7 pp of headroom, and specialty role prompts contribute $+0.37$ pp.

A nine-member panel contains 1.27 independent opinions. Across Independent, Centralized and Decentralized at $n_a \geq 3$, the mean pairwise er-

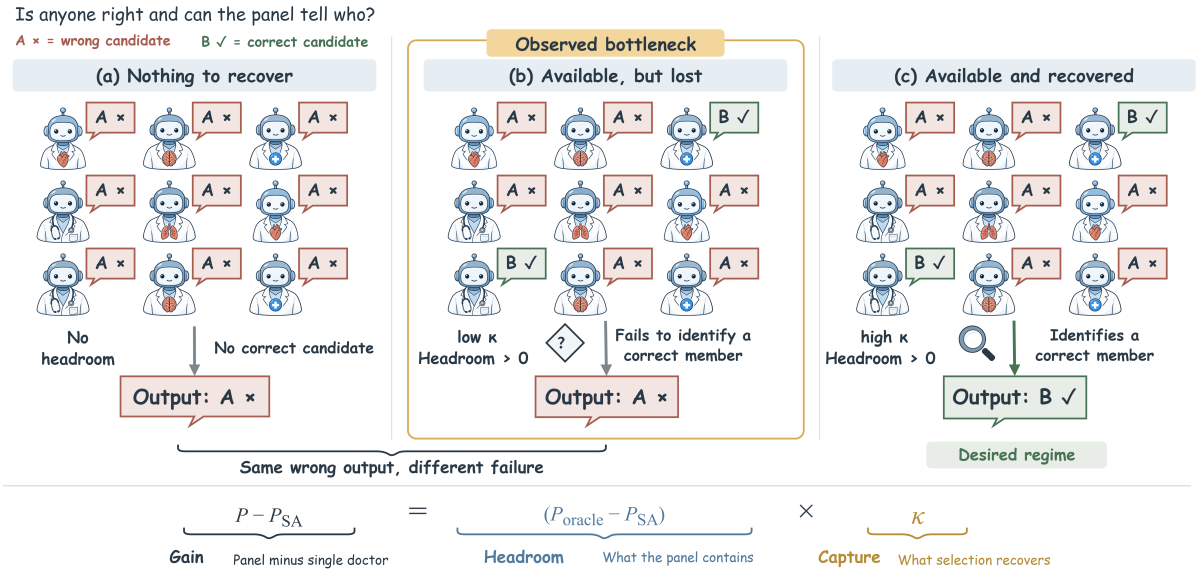


Figure 3: **A panel can fail two ways, and accuracy does not tell them apart.** (a) No correct answer leaves no headroom. (b) A correct first-round answer is lost, the observed bottleneck with low κ. (c) The correct answer is recovered with high κ. Eq. 3 gives gain as headroom (§4.3) times capture rate κ (§4.4).

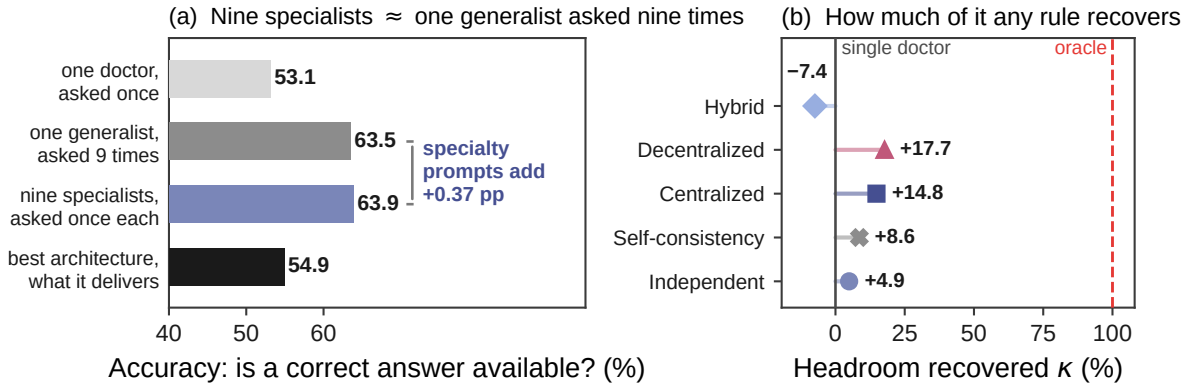


Figure 4: **A panel holds answers it cannot deliver.** (a) Bars compare single-doctor accuracy, oracle ceilings at $N = 9$ and panel accuracy. (b) Capture rate κ is pooled over configurations at $N \geq 3$: total gain divided by total headroom. Zero marks the single-doctor baseline; 100 marks an oracle selecting a correct answer whenever a member holds one.

ror correlation is $\varphi = 0.758$. These architectures reuse the same first opinions (§3.2), so their first-round correlations coincide by design. Hybrid is reported separately because its φ is measured only on escalated items (§4.4). By Eq. 1 this leaves $N_{\text{eff}} = 1.19$ at N=3, 1.24 at N=5 and 1.27 at N=9 (Table 3). Tripling the panel adds 0.08 of an independent opinion. Panel size alone predicts gain at cross-validated $R^2 = -0.007$ over the 225 OpenAI configurations, worse than predicting the mean (§G).

Discussion drives the effective panel size to one. Across the 96 Decentralized configurations, peer discussion raises φ from 0.758 (SD 0.117) before the exchange to 0.990 (SD 0.010) after it (Fig. 5). At $\varphi = 0.990$ a nine-agent panel has $N_{\text{eff}} = 1.01$.

What discussion changes. Across 42,464 (item, panel position) pairs, 37.4% of opinions change between the first round and the last. Of the changes, 39.0% go from wrong to right, 21.4% from right to wrong, and 39.6% from one wrong answer to a different wrong one. Three-fifths of all movement ends somewhere other than the correct option. The net effect on individual opinions is positive (2,787 more corrections than corruptions, or +6.6 pp of positions), while the members' error correlation rises.

Table 1: **Main results.** Each architecture appears at its best panel size; gain compares the best with the single doctor. Bold accuracies mark the row's best architecture; bold gains mark gains of at least 5 pp in absolute value.

	Single	Best multi-agent variant (accuracy % / n_a)				Gain
Model	doctor	Independ.	Central.	Decentr.	Hybrid	best
MedXpertQA — expert-level reasoning, 10 options						
gpt-4.1-nano	15.6	16.0 / 9	**19.2** / 1	16.8 / 5	18.0 / 9	+3.6
gpt-5-nano	32.0	35.6 / 5	**38.4** / 3	38.0 / 7	35.2 / 5	**+6.4**
gpt-5-mini	41.6	47.2 / 5	**49.2** / 3	48.0 / 5	45.2 / 1	**+7.6**
gemini-3.5-flash-lite	34.8	36.8 / 5	38.4 / 5	**39.2** / 7	36.4 / 9	+4.4
gemini-3.7-flash	60.4	62.8 / 3	**64.0** / 3	**64.0** / 3	61.6 / 3	+3.6
claude-haiku-4.5	23.2	22.0 / 9	22.8 / 1	**23.2** / 7	22.8 / 1	+0.0
claude-sonnet-5	49.6	47.2 / 3	**48.8** / 1	47.2 / 5	47.2 / 1	-0.8
deepseek-v4-flash	37.2	42.0 / 9	**44.4** / 3	**44.4** / 5	36.0 / 3	**+7.2**
MedAgentsBench-hard — items where strong models still fail						
gpt-4.1-nano	20.4	18.0 / 1	**21.6** / 7	18.4 / 5	16.0 / 9	+1.2
gpt-5-nano	32.4	**34.8** / 5	34.4 / 3	34.4 / 9	34.0 / 5	+2.4
gpt-5-mini	44.4	44.8 / 9	46.0 / 7	**46.8** / 9	46.0 / 1	+2.4
gemini-3.5-flash-lite	31.2	34.4 / 3	34.8 / 3	**35.2** / 3	33.2 / 7	+4.0
gemini-3.7-flash	56.4	57.2 / 9	57.2 / 9	58.0 / 9	**58.8** / 9	+2.4
claude-haiku-4.5	20.8	21.6 / 9	**25.6** / 3	22.4 / 5	22.4 / 7	+4.8
claude-sonnet-5	47.6	45.2 / 9	45.2 / 7	**45.6** / 9	**45.6** / 7	-2.0
deepseek-v4-flash	39.6	39.2 / 9	40.4 / 7	**40.8** / 7	32.0 / 9	+1.2
MedQA (USMLE) — licensing-exam questions, 4 options						
gpt-4.1-nano	66.0	67.2 / 9	64.0 / 9	**67.6** / 5	65.6 / 9	+1.6
gpt-5-nano	88.0	**89.6** / 5	**89.6** / 5	**89.6** / 9	88.8 / 5	+1.6
gpt-5-mini	91.6	92.4 / 1	93.2 / 1	92.8 / 5	**93.6** / 1	+2.0
gemini-3.5-flash-lite	84.8	87.2 / 9	**89.6** / 3	89.2 / 7	87.6 / 9	+4.8
gemini-3.7-flash	96.4	**96.4** / 7	**96.4** / 3	**96.4** / 3	96.0 / 9	+0.0
claude-haiku-4.5	79.6	80.4 / 9	**81.6** / 5	81.2 / 3	79.2 / 9	+2.0
claude-sonnet-5	94.0	93.2 / 9	94.4 / 9	**94.8** / 5	94.0 / 5	+0.8
deepseek-v4-flash	88.0	93.2 / 7	**94.4** / 9	**94.4** / 7	88.8 / 7	**+6.4**

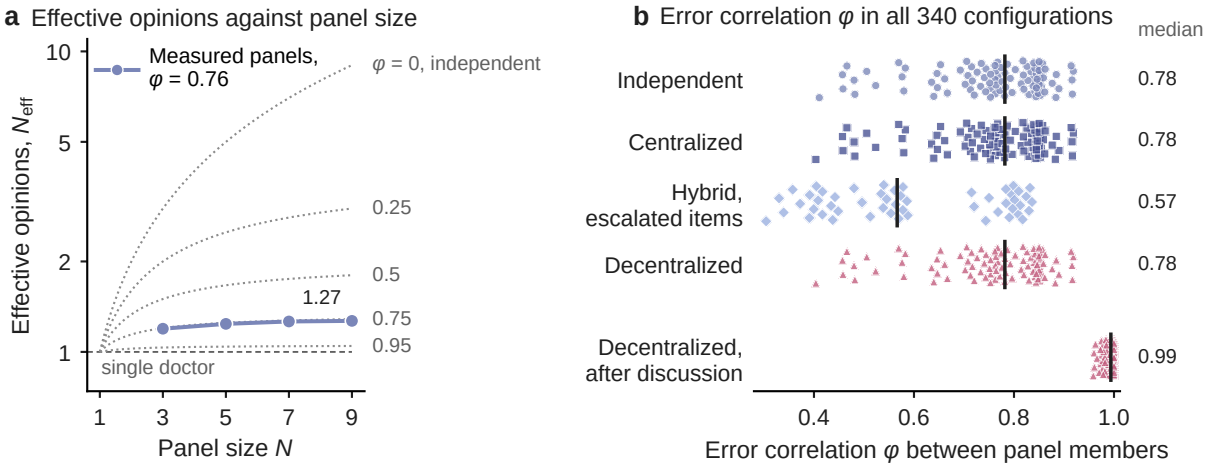


Figure 5: **Adding agents adds almost no information, and discussion removes what is left.** (a) N_{eff} uses logarithmic vertical scaling against panel size N. Dotted curves fix φ; the dashed line marks one opinion (single doctor). (b) Points show first-round error correlation φ for 340 configurations by architecture and last-round φ for 96 Decentralized configurations. Ticks mark medians; Hybrid uses escalated items.

Role prompting matters most without communication. We measure how much accuracy gain specialty role prompts contribute by comparing specialist panels with the GENERIC PANEL condition. Arms share the model, the temperature, the architectures and the same 250 items (gpt-5-nano, MedXpertQA), and we pair them over the 12 matched (architecture, $N \geq 3$) con-

Table 2: **Gain decomposed.** Collaboration gain is headroom times capture rate (Eq. 3). n is configurations.

P_{SA} band	n	headroom	capture rate κ	gain
$< 25\%$	56	+6.7 pp	-4.4%	-0.29
25–50%	144	+14.3 pp	$+6.6\%$	$+0.95$
50–70%	44	+6.3 pp	$+10.7\%$	$+0.68$
$> 70\%$	92	+4.1 pp	$+29.5\%$	$+1.20$

Table 3: **Effective independent opinions.** φ is the mean pairwise error correlation between panel positions; N_{eff} follows Eq. 1. The shaded Hybrid row is measured only on the items its triage gate escalates.

Architecture	N	φ	N_{eff}	N_{eff}/N
Growing one panel				
Independent	3	0.756	1.19	40%
Independent	5	0.758	1.24	25%
Independent	9	0.762	1.27	14%
Architectures at N=9 share first opinions				
Centralized	9	0.762	1.27	14%
Decentralized	9	0.762	1.27	14%
Hybrid	9	**0.604**	**1.54**	17%

figurations. Against the single doctor at 32.0%, the specialist panel gains +4.63 pp. The Generic panel supplies +2.77 pp (SD 2.0 across the 12 configurations), and specialty role prompts supply +1.87 pp (SD 1.4). The role prompts' share of the gain is 81% under independent voting, 40% under a central orchestrator and 22% under discussion. Independent generic internist panels gain +0.50 pp averaged over $N = 3$–9.

Specialty matching has a negative mean effect across models. Every MedXpertQA item is labelled with the body system it concerns, and the router ranks specialties by relevance, so at a given panel size some items receive a panel containing the corresponding specialty (a neurologist on a nervous-system case) and others do not. We compare these groups to measure the effect of specialty matching.

We compare the *gain over the single doctor* on the same items, which absorbs item difficulty into the baseline. Across eight models at $n_a \in \{3, 5\}$, matching the specialist to the case changes gain by -1.27 pp on average across 16 contrasts, ranging from -12.5 to $+11.9$ pp.

Normalising correlation reduces the accuracy-gap coefficient. Training families are generations within one vendor, such as `gpt-4o`, `gpt-4.1` and `gpt-5`. We regress the 360 model–benchmark pairs in Table 4 on family identity and accuracy gap, using the normalisation defined in §3.1. On raw φ the accuracy-gap coefficient is $\hat{\beta} = -0.725$ ($R^2 = 0.328$). On $\varphi/\varphi_{\max}$ it is -0.034 ($R^2 = 0.015$), while the family term remains positive ($+0.107 \rightarrow +0.088$).

Mixing training families reduces correlation more than mixing vendors. Table 4 compares the four sources of panel diversity. Specialty prompts give $\varphi/\varphi_{\max} = 0.792$. Different checkpoints from one family give 0.561, and different training families within one vendor give 0.465. The latter yield 2.60 effective opinions in a nine-member panel, computed from raw φ. The 300 cross-vendor model–benchmark pairs over the same 750 items span six vendors on two continents (OpenAI, Google, Anthropic, DeepSeek, Alibaba and Zhipu). They give $\varphi/\varphi_{\max} = 0.504$ and $N_{\text{eff}} = 2.29$ on raw φ. The same-vendor coefficient on normalised correlation is $\hat{\beta}_{\text{same vendor}} = -0.032$, with the family term at $+0.088$. Restricted to pairs whose error rates differ by less than 5 pp, where $\varphi_{\max}$ averages 0.9, cross-vendor pairs give 0.480 against 0.386 within a vendor (2.02 against 2.31 effective opinions on raw φ).

Accuracy gaps correlate with raw error correlation. Across the 360 model–benchmark pairs, the correlation between accuracy gap and raw error correlation is $r = -0.550$. Pairs within 5 pp of each other have mean $\varphi = 0.438$, giving 2.00 effective opinions for nine members by Eq. 1. Pairs more than 30 pp apart have mean $\varphi = 0.156$, giving 4.00 effective opinions.

After normalisation by Eq. 2, the correlation with accuracy gap is $r = -0.043$.

A panel of one strong, one mid and one weak specialist has lower accuracy than three strong members under both Centralized and Decentralized on MedXpertQA (§I).

Most headroom requires choosing a member per item. Two oracles separate fixed specialty selection from per-item selection. Always trusting whichever specialty position turns out best over the whole benchmark, a post-hoc choice no deployment could make in advance, reaches 54.6% against 53.1% for the single doctor. Choosing a correct opinion item by item reaches 63.9%. Of the 10.7 pp of headroom, 1.5 comes from picking a better specialist and 9.3 requires changing which member is selected from case to case.

Table 4: **Four sources of panel diversity.** Normalised correlation $\varphi/\varphi_{\max}$ accounts for unequal error rates. Rows vary prompts, checkpoints, training families and vendors; n counts model–benchmark pairs. N_{eff}^{9} is computed from raw φ.

Source of diversity	n	φ	$\varphi_{\max}$	$\varphi/\varphi_{\max}$	N_{eff}^{9}
One model, different specialty prompts	—	0.758	0.958	0.792	1.27
Different checkpoints, one training family	21	0.479	0.854	0.561	1.86
Different families, one vendor	39	0.308	0.663	0.465	2.60
Different *vendors* (OpenAI/Google/Anthropic/DeepSeek/Alibaba/Zhipu)	300	0.366	0.726	**0.504**	**2.29**
fully independent	—	0	1	0	9.00

4.4 Can the room tell who? The capture rate

The capture rate κ, averaged over all configurations at $n_a \geq 3$, is 17.7% for peer discussion, 14.8% for an auditing orchestrator, 8.6% for budget-matched self-consistency and 4.9% for majority vote. Hybrid has $\kappa = -7.4\%$, below the single-doctor baseline (Fig. 4b). Four-fifths of the headroom is discarded by the best mechanism we test, and choosing the best architecture for each cell after the fact, which no deployment can do, reaches $\kappa = 23.3\%$ using the same estimator as the per-architecture values. At n_a=9 the best measured architecture scores 54.9%.

Disagreement includes pairs of wrong answers. Two specialists in the same panel give different answers on 14.1% of items before any exchange (MedXpertQA 19.6%, MedAgentsBench 17.1%, MedQA 5.6%). The corresponding rate is 21% for second opinions that produce a distinctly different diagnosis in a large second-opinion audit (Van Such et al., 2017). Across all pairs, 48.7% agree and are right, 37.2% agree and are both wrong, 8.4% split with exactly one right, and 5.7% split with both wrong. Of every disagreement, 40.2% is two agents reaching different wrong answers.

Error amplification locates the loss inside the episode. Error amplification A_e is 0.32 for Centralized, 0.29 for Independent, 0.25 for Decentralized and 0.09 for Hybrid.

Mean rationale overlap among first opinions is 0.383 for pairs that are both right, 0.346 for pairs that are both wrong and 0.199 for pairs with one right and one wrong answer. Overlap differs by 0.15–0.18 between pairs with the same correctness and pairs with mixed correctness, and by 0.037 between both-right and both-wrong pairs. Similar rationales occur when both members are right and when both are wrong.

First-round agreement signals weakly distinguish correct answers. Across $71{,}793$ panel episodes at $N \geq 3$, first-round answer entropy predicts panel correctness at AUC 0.592 and Youden $J = 0.177$. Majority share reaches AUC 0.591 and $J = 0.177$. Members' errors correlate at $\varphi = 0.76$, and panels agree both on correct answers and on shared errors.

A selector trained on the answers does not beat counting votes. For each of the $71{,}793$ panel episodes we formed one row per distinct candidate answer, with fifteen features computable from the panel's own first round: vote share, margin over the runner-up, the mean, maximum, minimum and spread of the confidences attached to that candidate, that candidate's confidence relative to the panel mean, rationale lengths, the number of distinct answers, the entropy of the vote distribution, and panel size. The label is whether the candidate is correct. We trained a logistic regression and a gradient-boosting classifier under five-fold cross-validation grouped by item: fitted on ground truth and tested in the same distribution using these fifteen features.

Neither beats majority vote. Selecting the highest-scoring candidate gives 53.55% (logistic) and 53.37% (boosted trees) against 53.65% for taking the plurality, while the oracle reaches 61.95%. Rescaled so that plurality scores 0 and the oracle 100, the two learned selectors score -1.3 and -3.4. Neither classifier recovers any of the 8.30 pp between plurality and oracle using these fifteen features.

Hybrid escalates on one-fifth of items. Hybrid escalates on 20% of items overall. The rate varies across model–benchmark cells: 76.8% for `claude-sonnet-5` on MedXpertQA, 0.0% for `gpt-4.1-nano` on MedQA, where the gate never once fires in $1{,}000$ episodes (its 250 items at four panel sizes).

Table 5: **Two published frameworks, measured the same way.** Both run on `gpt-4.1-nano` over the same 750 items, with the single-doctor baseline from the main grid. N_{eff} is stated at nine agents so the rows are comparable.

	Gain	φ	N_{eff}
Ours, `gpt-4.1-nano`	+2.1	0.792	1.23
MedAgents	+5.7	0.860	1.14
MDAgents	+1.2	0.863	1.14

Per benchmark, φ ranges 0.803–0.890 over all six framework–benchmark combinations.

Table 6: **Clinical safety.** At N=9, unan. is the unanimity rate, $P(\text{wr.} \mid \text{un.})$ is the error rate of a unanimous panel, and $P(\text{wr.} \mid \text{div.})$ is the error rate of a divided panel. Red marks the values discussed.

Architecture	unan.	$P(\text{wr.} \mid \text{un.})$	$P(\text{wr.} \mid \text{div.})$
Independent	67.5%	38.1%	63.7%
Centralized	67.0%	38.1%	62.0%
Decentralized	**98.0%**	44.9%	71.9%
Hybrid†	51.2%	49.4%	61.8%

†Hybrid is measured on the 20% of items its triage gate escalates.

4.5 The same measurement on two published frameworks

To measure systems we did not design, we ran MedAgents (Tang et al., 2024) and MDAgents (Kim et al., 2024) on the same 750 items, from the prompts in their own repositories rather than from the papers' descriptions (§J).

The frameworks differ in routing, discussion and aggregation. MedAgents partitions each question into five clinical domains and two option domains on the fly, has each domain expert write an analysis, synthesises them into one report, and then puts the report to a vote, revising it until no expert objects. MDAgents first classifies the question as basic, intermediate or advanced, answers the basic ones with a single agent, and for the rest recruits a panel and lets its members choose whom to talk to, round by round, until nobody wants to speak. Peer context also differs: a MedAgents voter reads a report of roughly 8,000 characters, whereas our members share 900.

The frameworks have similar error correlations. MedAgents gains +5.7 pp over the single doctor and MDAgents +1.2 (Table 5). Their error correlations are 0.860 and 0.863. Across all six framework–benchmark combinations φ stays between 0.803 and 0.890, on baselines running from 15.6% to 66.0%, and both frameworks are more correlated than our `gpt-4.1-nano` panels at $\varphi = 0.792$. Six experts amount to 1.1 independent opinions; nine members of our `gpt-4.1-nano` panels amount to 1.23.

Revision ends early; complexity tracks difficulty on one benchmark. MedAgents revises its report until no expert votes against it. That loop reaches agreement on 96–99% of items and does so *in the first round* on 92–96%. Its revision loop stops when no expert objects, with no check on correctness. MDAgents routes on a complexity label that tracks difficulty only on MedXpertQA, where the items it calls most complex score 14% against 18 and 21. On MedAgentsBench its three levels differ by at most five points, while on MedQA the basic tier scores lowest (59 against 74 and 70).

4.6 Clinical safety: what consultation does to agreement

We measure panel unanimity and the error rates of unanimous and divided panels (Table 6).

Discussion raises unanimity and its error rate. At N=9, unanimous panels are wrong on 38.1–49.4% of the items they agree on; divided panels are wrong on 61.8–71.9%. Discussion raises unanimity from 67.5% under Independent to 98.0% under Decentralized, while the error rate given unanimity rises from 38.1% to 44.9%.

4.7 What a panel costs

Centralized is the least token-efficient architecture. Successes per 1,000 tokens: Hybrid 0.75, Independent 0.39, Decentralized 0.34, self-consistency 0.25, Centralized 0.20. Centralized is 3.7× less token-efficient than Hybrid, the most efficient condition.

On MedXpertQA at `gpt-5-nano`, Decentralized at N=3 reaches 38.0% while self-consistency at the closest matched dollar cost reaches 34.8%. On MedQA at the same tier, self-consistency at k=15 reaches 88.4% and costs three times as much as the best panel, which reaches 89.6%. On MedAgentsBench at T1, self-consistency (20.0%) trails Centralized (21.6%).

What a point of accuracy costs. Table 7 reports accuracy gain per dollar of additional spend for each cell's best configuration. Among configurations costing more than the single doctor, the cost multiple ranges from 1.9× to 11.5×, and gain per

Table 7: **Accuracy per dollar.** For each cell, the best configuration of any architecture and panel size: its gain over the single doctor, the additional spend per 1,000 questions, the cost multiple, and the ratio.

Benchmark	Tier	gain	extra $/1k	×cost	pp/$
MedXpertQA	T1	+3.6	0.08	1.9	**44.5**
MedXpertQA	T2	+6.4	0.84	5.1	7.6
MedXpertQA	T3	+7.6	1.76	3.5	4.3
MedAgentsBench	T2	+2.4	0.28	2.5	8.5
MedAgentsBench	T3	+2.4	7.12	11.5	0.34
MedAgentsBench	T1	+1.2	0.39	6.2	3.1
MedQA	T1	+1.6	0.27	4.2	6.0
MedQA	T2	+1.6	0.49	4.1	3.3
MedQA	T3	+2.0	−0.14	0.8	—

additional dollar varies by two orders of magnitude. It is 7.6 and 4.3 percentage points per additional dollar per thousand questions on the two MedXpertQA cells with the largest gains, against 0.34 on MedAgentsBench at T3, where a +2.4 pp gain costs 11.5×. On MedQA at T3 the best configuration is Hybrid at n_a=1, which costs less than the single-doctor baseline.

5 Conclusion

A panel of nine language-model specialists is worth one physician. Error correlation and capture rate κ identify two constraints on consultation gains.

Role-prompted specialists' errors correlate at $\varphi = 0.758$, leaving a nine-member panel with 1.27 effective independent opinions. Discussion raises φ to 0.990, leaving 1.01. Self-consistency supplies 97% of the 10.7 pp of headroom, while specialty role prompts add +0.37 pp to the oracle ceiling. Computed from raw φ, mixing model generations within one vendor yields 2.60 effective opinions out of nine, and mixing models from six vendors yields 2.29.

Decentralized recovers 17.7% of headroom and Centralized 14.8%. Independent recovers 4.9%, and Hybrid has $\kappa = -7.4\%$. Neither learned selector improves on plurality.

Discussion raises unanimity to 98.0% while 44.9% of unanimous answers are wrong (Decentralized, N=9; Table 6).

Before deploying a panel, compare it with the single doctor and budget-matched self-consistency on the intended case mix.

Making medical panels agree is easy; making them verifiable is the open problem.

6 Limitations

Our tasks are multiple-choice examination items, not clinical encounters. They carry no history taking, no test ordering, no longitudinal follow-up, and the exact-match endpoint rewards the option-selection component of diagnosis only. Results here do not extrapolate to tool-using clinical agents.

The diversity analysis is limited to evaluated models from six vendors (OpenAI, Google, Anthropic, DeepSeek, Alibaba and Zhipu; §4.3).

The specialty router ranks by relevance, so larger panels append progressively less relevant specialists. Panel size is therefore entangled with specialty relevance. The GENERIC PANEL control removes specialty roles by giving every member the same generic internist prompt, separating panel size from specialty relevance (§4.3).

Ethics Statement

This work uses only public benchmark items derived from licensing examinations and published case collections. No patient data, no human subjects and no clinical deployment are involved.

The findings identify a deployment risk: panels present shared errors as expert consensus (§4.6). Apparent expert consensus also poses a risk of medical misinformation that reads as authoritative.

Our findings motivate three mitigations. First, do not present unanimity as a trust cue in a clinical interface. Reporting "all five specialists agree" conveys agreement that requires separate verification. Second, show the panel's *disagreements* without resolving them into a single consensus. Third, route verification to retrieved evidence or a clinician. Zöller et al. (2025) attribute gains in human–model collective diagnosis to complementary errors.

We report the failure mode as a measured property of systems already being built and deployed, not as a novel capability.

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A Model capability index

Kim et al. (2026) score models on a published Intelligence Index and report that a *task-grounded* capability metric explains more variance ($R^2 = 0.413$) than the generic one ($R^2 = 0.373$). We define the task-grounded capability index I as a model's mean single-doctor chain-of-thought accuracy across the three benchmarks. Table 8 gives the components. The three OpenAI capability tiers span $I = 34.0$ to 59.2, and their single-benchmark baselines span 15.6% to 91.6%, crossing the 45% region from both sides.

B Domain complexity

B.1 Complexity metric construction

Domain complexity $D \in [0, 1]$ is the arithmetic mean of three complementary measures (Kim et al., 2026): the **performance ceiling** $1 - p_{\max}$, where $p_{\max}$ is the highest performance any evaluated system reaches; the **coefficient of variation** σ/μ of performance across all configurations, a scale-invariant measure of relative spread; and the **best-model baseline** $1 - p_{\text{best}}$, where p_{best} is the strongest single-model performance on the dataset.

B.2 Domain characterisation

Table 9 gives the components. MedQA is a low-complexity domain ($D = 0.060$); MedXpertQA and MedAgentsBench-hard have similar, higher complexity values ($D = 0.370$ and 0.392).

B.3 The critical threshold does not transfer

A critical complexity threshold at $D \approx 0.40$ has been reported for general-domain agents (Kim et al., 2026): below it multi-agent architectures yield net positive returns, above it coordination overhead consumes resources otherwise spent on reasoning. MedXpertQA ($D = 0.370$) and MedAgentsBench-hard ($D = 0.392$) differ by 0.022 in complexity yet by 1.8 pp in mean multi-agent gain ($+1.44$ pp versus -0.34 pp), giving opposite signs below the reported threshold. Gains also vary *within* each benchmark: on MedAgentsBench-hard the per-tier gains run -2.78, $+1.02$, $+0.66$ pp as P_{SA} moves

Table 8: **Capability index.** Entries are single-doctor CoT accuracies (%); bold I is their mean across benchmarks. The final column gives model settings.

Model	MedXpertQA	MedAgentsBench	MedQA	I (mean)	Setting
OpenAI — the capability ladder the grid is run on					
`gpt-4.1-nano`	15.6	20.4	66.0	**34.0**	—
`gpt-5-nano`	32.0	32.4	88.0	**50.8**	effort=low
`gpt-5-mini`	41.6	44.4	91.6	**59.2**	effort=low
Google — a second vendor, for the diversity ladder					
`gemini-3.5-flash-lite`	34.8	31.2	84.8	**50.3**	—
`gemini-3.7-flash`	60.4	56.4	96.4	**71.1**	—
Anthropic — a third vendor, for the diversity ladder					
`claude-haiku-4.5`	23.2	20.8	79.6	**41.2**	—
`claude-sonnet-5`	49.6	47.6	94.0	**63.7**	—
DeepSeek — a fourth vendor in the grid; Alibaba and Zhipu contribute single-doctor baselines to the diversity ladder					
`deepseek-v4-flash`	37.2	39.6	88.0	**54.9**	—
`deepseek-v4-pro`	44.8	42.0	92.4	**59.7**	—
`qwen3.8-flash`	34.4	32.8	88.8	**52.0**	—
`qwen3.8-max`	42.0	40.0	88.8	**56.9**	—
`glm-5.3-flash`	47.6	45.6	93.2	**62.1**	—
`glm-5.3`	42.8	46.4	93.6	**60.9**	—

Capability-matched cross-vendor pairs:
`gemini-3.5-flash-lite` (50.3) with `gpt-5-nano` (50.8), and
`deepseek-v4-pro` (59.7) with `gpt-5-mini` (59.2) across the Pacific.

Table 9: **Domain complexity of the three benchmarks.** Bold D averages the performance ceiling, relative spread and best-model baseline; mean multi-agent gain is in pp.

Benchmark	$1 - p_{\max}$	σ/μ	$1 - p_{\text{best}}$	D	mean MAS gain
MedXpertQA	0.360	0.353	0.396	**0.370**	+1.44 pp
MedAgentsBench-hard	0.412	0.328	0.436	**0.392**	-0.34 pp
MedQA (USMLE)	0.036	0.108	0.036	**0.060**	+0.72 pp

from 20.4% through 32.4% to 44.4%. The configuration-level baseline P_{SA} distinguishes these gains.

C Datasets

MedXpertQA-Text (Zuo et al., 2025): 2,450 expert-level items, ten options each, tagged by 12 body systems and 3 task types. We draw a stratified sample interleaved across (body system $\times$ task) so that any prefix is approximately balanced, then take the first 250. **MedQA-USMLE** (Jin et al., 2021): 1,273 four-option test items; randomly shuffled with a fixed seed, first 250. **MedAgentsBench-hard** (Tang et al., 2025): the contamination-screened hard split, 894 items after excluding the `medqa_5options` subset, stratified-interleaved across the ten source subsets, first 250. All samples are nested prefixes, so a larger sample re-uses every call of a smaller one.

Item difficulty is additionally tagged easy/medium/hard by the pass rate of three mid-tier models at k=5; on the MedXpertQA sample this yields 13/62/125 items.

D Additional results

Figure 6 plots the gains in Table 1; Figure 7 breaks the same grid out by vendor. Figures 8 and 9 give the full panel-size curves; gains mostly arrive by n_a=3. A panel larger than n_a=5 helps in 29 of 96 (cell, architecture) pairs, by at most 2.8 pp. Figure 10 shows performance distributions within and between architectures (§4.2).

E Implementation details

E.1 Technical infrastructure

All models are accessed through one OpenAI-compatible client with a content-addressed disk cache, exponential backoff with rate-limit-aware retry, a per-call JSONL ledger recording tokens, latency and cost, and a cross-process spend journal guarded by an OS file lock. Every LLM call is cached on the full request (model, system, messages, temperature, seed, max tokens, effort, JSON mode, tag), so a re-run is a replay.

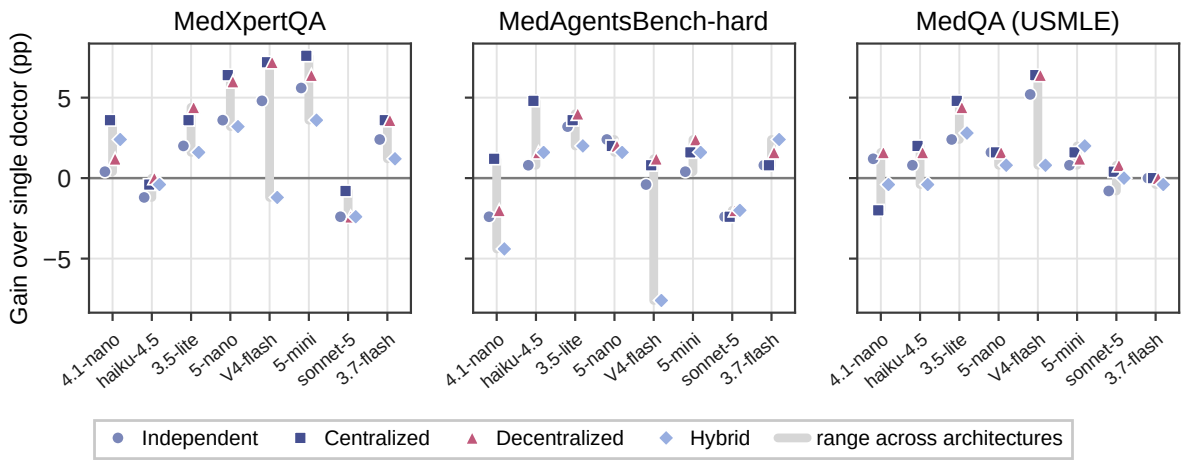


Figure 6: **Every architecture lands close to the single doctor.** Each panel shows a benchmark. The eight models are ordered by I, their mean single-doctor accuracy across benchmarks. Markers show each architecture's gain (pp) at its best panel size; grey bars span the four architectures. Figure 7 gives the vendor breakdown.

Table 10: **Model selection probe.** Eight candidates were screened on 40 items per benchmark with single-doctor CoT before the grid was run. Shaded rows mark the selected models.

Model	Setting	Role	MedXpertQA	MedQA	USD / 1k items
Selected					
`gpt-4.1-nano`	—	tier T1	25.0%	75.0%	0.09
`gpt-5-nano`	effort=low	tier T2	37.5%	95.0%	0.17
`gpt-5-mini`	effort=low	tier T3	55.0%	97.5%	0.64
`gemini-3.5-flash-lite`	—	vendor contrast	30.0%	95.0%	0.09
`gemini-3.7-flash`	—	vendor contrast	80.0%	97.5%	0.41
Screened, not selected (used only for the pairwise φ analysis)					
`gpt-4o-mini`	—	—	20.0%	87.5%	0.13
`gpt-4.1-mini`	—	—	32.5%	85.0%	0.41
`gpt-5.4-nano`	effort=low	—	47.5%	95.0%	0.14

E.2 Agent configuration

Single-doctor baselines answer in one call. Independent panels deploy N specialists with majority-vote aggregation, ties broken by mean stated confidence and then option order. Centralized systems use one orchestrator over N sub-agents with a maximum of 2 orchestration rounds and one re-task per round. Decentralized systems run N agents through up to 3 discussion rounds, stopping early only on unanimity. Hybrid systems combine a generalist triage gate with the specialist panel and escalate to full discussion when the panel has no majority. Agent temperature is 0.7 within panels, 0.3 for single-doctor baselines and for orchestrator/triage turns. Reasoning models ignore temperature and are called at `effort=low` with a minimum completion budget of 1,600 tokens, since reasoning tokens are drawn from the same budget and a smaller cap returns empty output.

E.3 Prompt compilation

Every agent receives the same rendered item (stem, then options as `(A) ...`) and the same answer schema: JSON with `answer`, `confidence` (0–100) and `reason` (at most 45 words). For specialist panel members, the system prompt names their specialty. Peer context in communicating architectures is capped at a fixed 900-character budget, balanced-truncated across peers and *independent of* N, so a panel-size effect cannot be a context-length effect. Peer ordering is deterministically rotated per (item, round) so that the first-listed speaker is decorrelated from the top-ranked specialty.

F Complete results grid

The per-configuration grid contains 480 multi-agent configurations and their single-model controls, with accuracy and cost for each. It is released with the code and the raw episodes used to

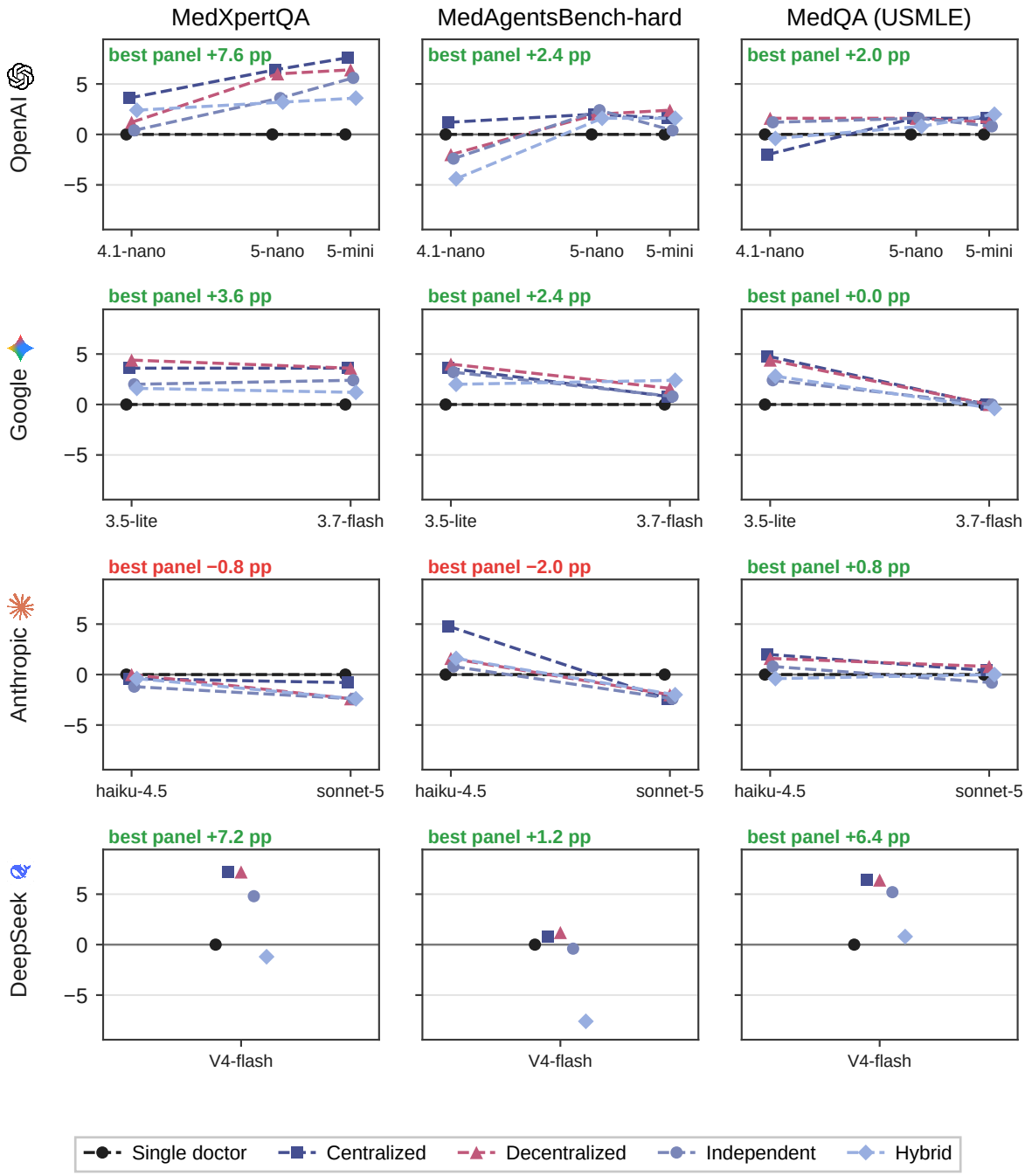


Figure 7: **The capability grid in full.** Figure 6's gains share one scale (pp) across vendor rows and benchmark columns; models are positioned by I. Black shows the single doctor at zero. Coloured annotations give each vendor's strongest model's best gain.

compute it.

G The published scaling form, fitted to medical data

The main text measures collaboration gain directly. We also fit the scaling form proposed for general-domain agent systems (Kim et al., 2026) to the OpenAI grid.

With accuracy as the response, the fit is degenerate. Fitting that form directly gives $R^2 = 0.997$ (5-fold cross-validated $R^2 = 0.996$), which is an artefact of our design rather than a result. The response is accuracy and one predictor is the single-doctor baseline P_{SA}; the same items and the same model underlie both terms, so P_{SA} alone nearly determines accuracy ($\hat{\beta}_{P_{\mathrm{SA}}} = 0.867$). We therefore refit with collaboration *gain* as the re-

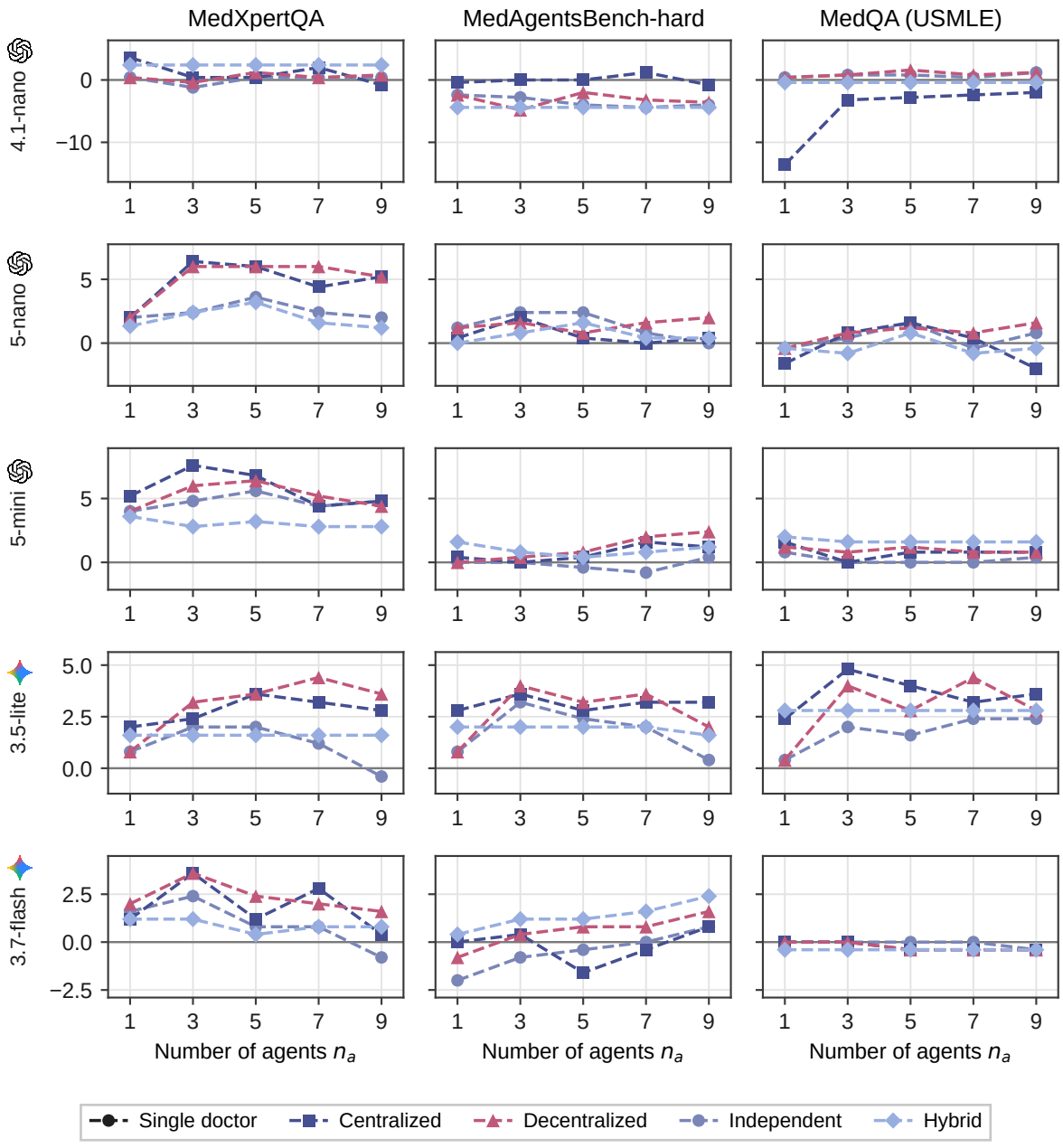


Figure 8: **Number-of-agents scaling, OpenAI and Google.** Gain over the single doctor (pp) is plotted against n_a for each architecture. Rows are models, columns are benchmarks, and each row shares a vertical scale. Figure 9 covers Anthropic and DeepSeek.

sponse.

Fitting collaboration gain. With gain $= P -$ P_{SA} as the response, $R^2 = 0.528$ and 5-fold cross-validated $R^2 = 0.440$. The coefficients below are fitted to $n = 225$ OpenAI configurations, including the self-consistency control.

term	$\hat{\beta}$
$\log(1 + A_e)$	-0.150
R (redundancy)	$+0.084$
c (message density)	$+0.006$
$\log(1 + O\%)$	$+0.005$
$P_{\text{SA}} \times \log(1+n_a)$	$+0.029$
$\log(1 + n_a)$	-0.005
P_{SA}	-0.113
P_{SA}^2	$+0.031$

Error amplification has a negative coefficient, while redundancy, message density and overhead have positive coefficients.

$T_{\text{tools}} = 0$ throughout, so the $\log(1 + T)$ main

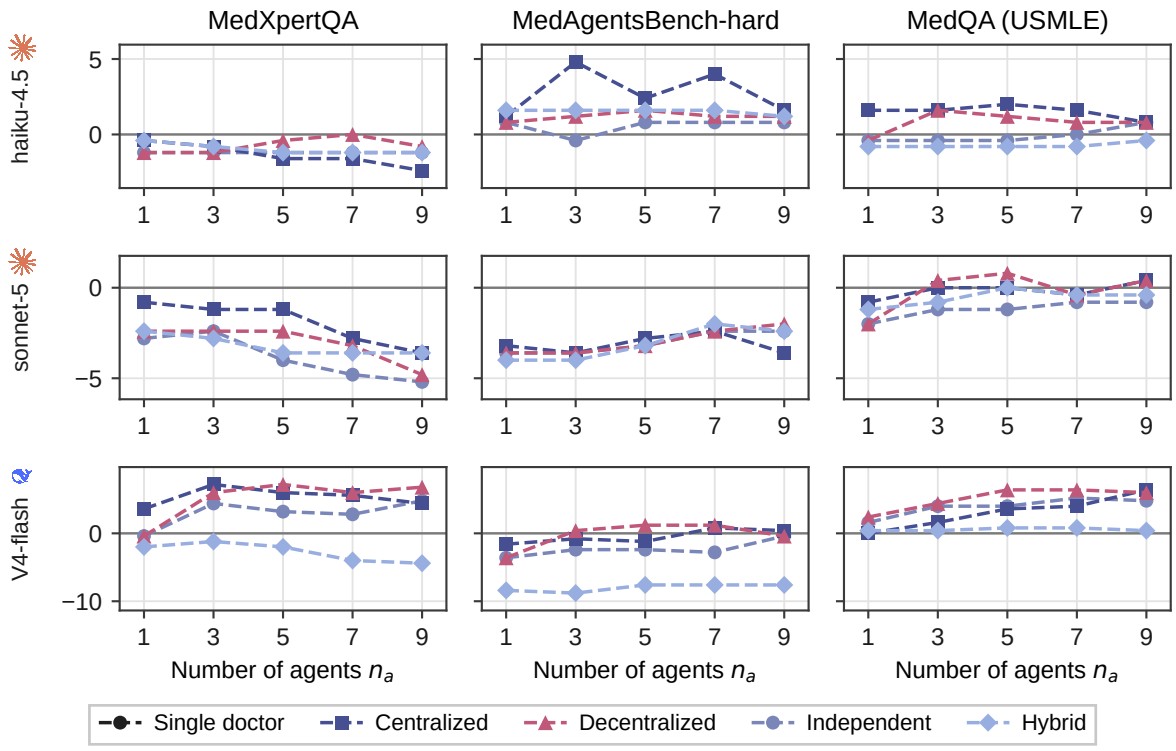


Figure 9: **Number-of-agents scaling, Anthropic and DeepSeek.** As Figure 8, for the remaining three models.

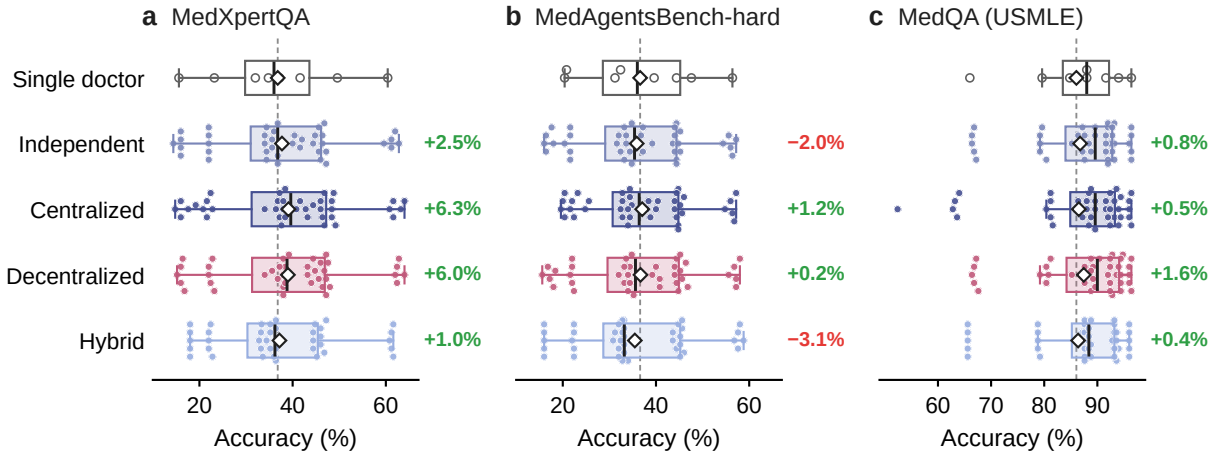


Figure 10: **Single doctor versus panel: performance distributions.** Horizontal boxes show accuracy by architecture and benchmark: Single doctor, Independent, Centralized, Decentralized and Hybrid. Each point is a configuration. Boxes show medians and interquartile ranges (IQR); whiskers extend to 1.5× IQR; diamonds mark means. Dashed lines mark single-doctor means; numbers give changes in mean accuracy as percentages of those means.

effect and the $O\% \times T$ and $E_c \times T$ interactions vanish identically.

Robustness. Capability in this regression is each tier's MedQA probe accuracy, which differs from I defined above. The linear and quadratic capability coefficients have opposite signs ($+0.302$ and -0.078). With three capability tiers the quadratic is unidentifiable and we do not interpret it.

Regularised alternatives do not improve prediction: Lasso (10-fold CV for λ) retains 9 of 11 predictors at CV $R^2 = 0.403$ and Ridge reaches 0.165, both below the unregularised 0.440, so we retain the full specification.

In nested model comparisons (5-fold CV R^2), the full model reaches 0.440 and capability alone 0.206. Coordination metrics alone reach 0.095, the single-doctor baseline alone 0.051 and panel size alone -0.007, worse than predicting the

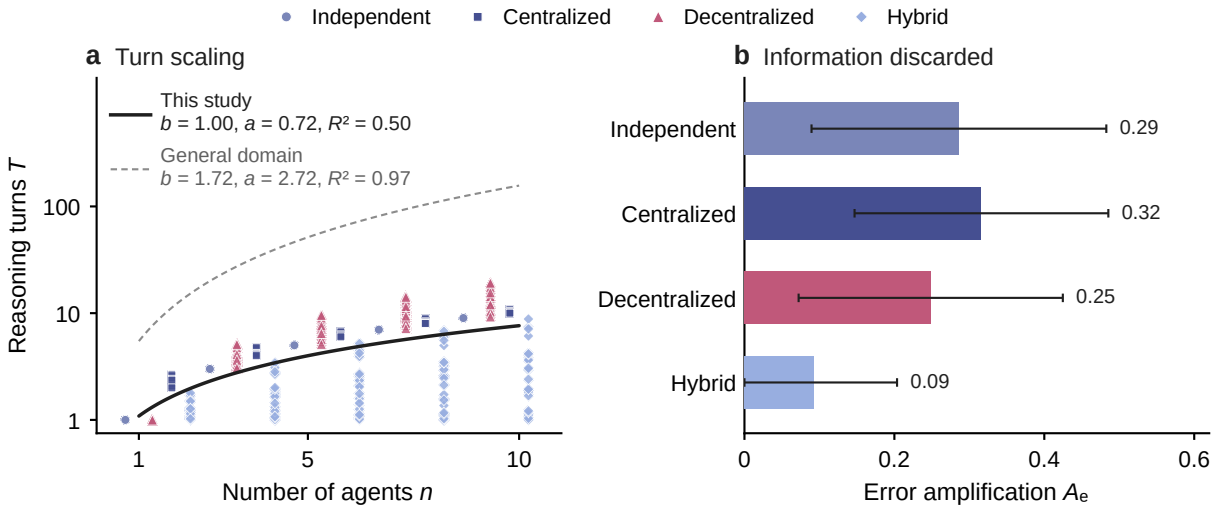


Figure 11: **A panel discards what it holds.** (a) Multi-agent configurations plot turns T against n (orchestrator included). The solid line fits $T = a(n + 0.5)^b$; dashes give the general-domain law. (b) Bars show mean $\pm$ SD across configurations by architecture; lower whiskers truncate at zero ($A_e \geq 0$). A_e measures episodes losing correct first-round answers to wrong outputs, normalised by first-round agents' error rate.

mean.

H Coordination dynamics

Figure 11 plots turn counts and error amplification.

Turns grow linearly; message density explains little variance. Fitting $T = a(n + 0.5)^b$ to the 480 multi-agent configurations gives $a = 0.72$, $b = 1.002$, $R^2 = 0.499$; the fitted exponent means turns grow in proportion to agents. The dashed general-domain law is $T = 2.72(n + 0.5)^{1.72}$, $R^2 = 0.97$ (Kim et al., 2026). Mean turn counts by architecture are Decentralized 7.1, Centralized 6.2, Independent 5.0, Hybrid 2.2; the single doctor takes exactly 1.0 turn in every cell.

$S = 0.510 - 0.026 \ln(c)$, $R^2 = 0.021$: message density explains almost none of the variance in accuracy. Our Centralized panels run at $c = 0.87$ and Decentralized at $c = 0.68$, above the $c^* = 0.39$ plateau reported by Kim et al. (2026).

Coordination failure needs a channel to occur on. We classify first-round errors into four categories, following the multi-agent failure taxonomy of Cemri et al. (2025):

	Indep.	Centr.	Decentr.	Hybrid
Logical contradiction	5.4%	5.4%	5.4%	3.2%
Numerical drift	5.8%	5.8%	5.8%	4.0%
Context omission	13.8%	13.8%	13.8%	7.5%
Coordination failure	**0.0%**	2.1%	0.8%	0.0%

Coordination failure is 0.0% under Independent coordination: with $C = \emptyset$ there is no channel on which it could occur. Hybrid is also at 0.0%. Kim et al. (2026) report that peer verification cuts logical contradiction from 16.8% to 11.5%; our first-round contradiction rate is 5.4% across communicating and non-communicating architectures. Numerical drift is rarer than in tool-using tasks (4–6% against 19–26%).

I Where capability belongs inside a panel

A separate arm varies the capability of the panel's members (Fig. 12).

Specialist capability matters four times more than the orchestrator's. Replacing the whole panel with weak models reduces accuracy by 34.4 pp (49.2% to 14.8%). Splitting that substitution isolates where the loss comes from. Downgrading only the *orchestrator*, keeping strong specialists, reduces accuracy by 3.6 pp, so the panel retains 90% of the homogeneous-high span. Downgrading only the *specialists*, keeping a strong orchestrator, reduces accuracy by 14.8 pp, retaining 57%. The asymmetry is 4.1×.

A weak orchestrator overrides a correct specialist majority. The largest single degradation anywhere in this study is a Centralized panel with a weak model on MedQA at $P_{\text{SA}} = 66.0\%$: −13.60 pp. In those episodes the orchestrator overrides a correct specialist majority.

Mixed-capability panels score below homogeneous strong panels. A panel of one strong, one

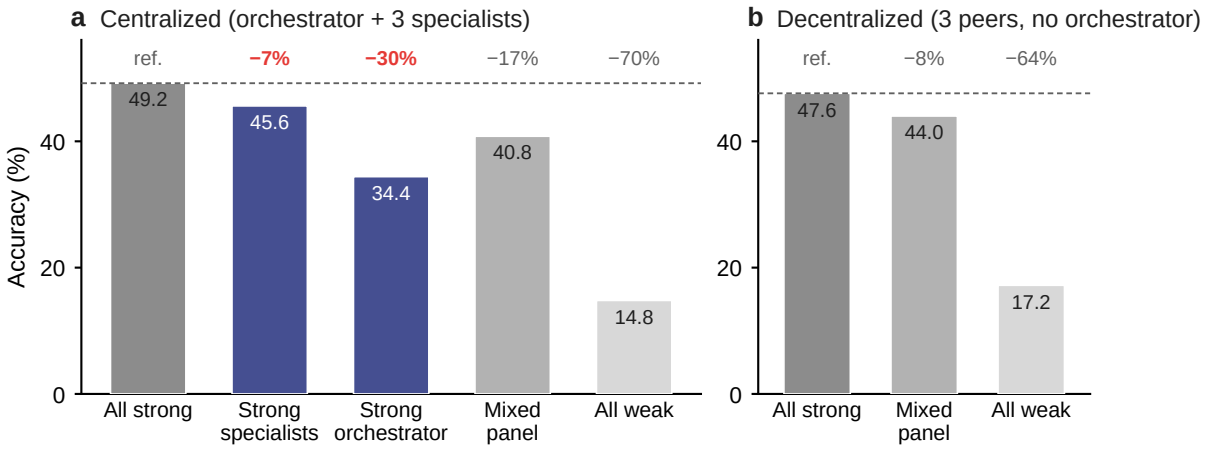


Figure 12: **Agent heterogeneity.** Bars show one run's accuracy on 250 MedXpertQA questions at $N = 3$. Strong is `gpt-5-mini`; weak is `gpt-4.1-nano`; mixed panels also use `gpt-5-nano`. *Strong specialists* pairs a weak orchestrator with strong specialists; *strong orchestrator* reverses this. Decentralized has no orchestrator. Dashed lines mark all-strong panels; top-row percentages give relative changes.

mid and one weak specialist reaches 40.8% under Centralized and 44.0% under Decentralized, against 49.2% and 47.6% for homogeneous-high. Both sit below the strong homogeneous panel and well above the weak one, roughly where a capability-weighted average would put them. In one model family, Kim et al. (2026) report that a low-capability orchestrator with high-capability sub-agents outperformed a homogeneous high-capability panel by 31%. In our data that configuration is the second-best Centralized cell but still 7% below homogeneous-high, directionally matching what they observe for OpenAI models, where heterogeneous Centralized configurations degrade.

J Reproducing the two published frameworks

Both frameworks are implemented from their authors' code, not from the papers' prose. MedAgents' twelve prompt functions are copied verbatim from `prompt_generator.py` in `gersteinlab/MedAgents` and its control flow follows `fully_decode()`'s `syn_verif` branch, including `max_attempt_vote`=3 and the `cleansing_syn_report` step that prepends the question and options to the synthesised report. MDAgents follows `determine_difficulty` and `process_intermediate_query` from `mitmedialab/MDAgents`, with the three-level complexity prompt quoted exactly and the early stop that ends a round when no agent wishes to address another.

We deviate from the official code in three ways. MDAgents runs two turns per round against the official five, which reduces cost by roughly a factor of four and does not touch the first-round opinions φ is computed on. Both frameworks run on `gpt-4.1-nano`; their official prompts allocate 30 to 50 tokens to short outputs such as a yes/no vote, and a reasoning model spends that budget entirely on thinking and returns nothing, so a non-reasoning model is both workable and closer to the GPT-3.5 and GPT-4 the original papers used. MedAgents' published pipeline produces domain *analyses* rather than option choices at the point where we need per-expert answers, so each expert is additionally asked for a letter; that call sits outside the official control flow and does not affect the framework's output.

Answers are parsed by the explicit formats the prompts request. A fallback that accepted a bare capital letter had to be removed: MedXpertQA has ten options, so `I` is a legal answer and a sentence beginning "I . . . " was read as one. Raw responses are stored with every episode so a parsing question can be settled without rerunning.